Sandy Zhang

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Research Interests

HCI, with a focus on how interactive systems shape the ways people think, learn, communicate, and collaborate. I'm particularly interested in human-AI interaction, intentional technology use, and designing interfaces that help people remain engaged with their work, learning, and interactions with others as AI becomes increasingly integrated into everyday life.

Education

Princeton University **Sep. 2022 – May 2026**
B.S.E. in Computer Science, GPA: 3.7 | Minor in Finance *Princeton, NJ*

- **Honors:** *cum laude*
- **Senior Thesis:** *Email Digests as an Interface for Intentional Social Media Consumption*
  Advised by Prof. Andrés Monroy-Hernández, Princeton HCI Lab.

Publications & Presentations

Peer-Reviewed

1. Romina Mahinpei, Diya Ajay Hundiwala, **Sandy Zhang**, Lana Glisic, and Andrés Monroy-Hernández. *How Children Collaborate within Programmable AR Environments with Co-Located Collaborative Features.* Poster, ACM Conference on Computer-Supported Cooperative Work and Social Computing (**CSCW 2026**). arXiv:2608.11442

Manuscripts in Preparation

1. **Sandy Zhang**, Romina Mahinpei, and Andrés Monroy-Hernández. *Email Digests as a Delivery Interface for Intentional Social Media Consumption.* Manuscript in preparation for *ACM CHI Conference on Human Factors in Computing Systems (Extended Abstracts).*

Posters & Research Reports

1. Arjun Shetty, Brian Zhou, Kaden Kram, Karen Wu, **Sandy Zhang**. *On-Chain Blackjack: Verifiable Fairness in Decentralized Games.* Poster presented at the **Princeton University DeCenter Annual Spring Conference**, Spring 2025.
2. **Sandy Zhang**. *Comparative Analysis of Computational gRNA Design Tools for CRISPR.* Independent Research Report advised by Prof. Yuri Pritykin, Princeton University, Fall 2024.

Research Experience

Princeton HCI Lab **Aug. 2025 – Present**
Undergraduate Researcher, advised by Prof. Andrés Monroy-Hernández *Princeton, NJ*

- **Email digests as an alternative to algorithmic feeds.** Designed and built a digest layer on top of **Bonsai** that delivers **Bluesky** content as **finite, periodic newsletters**, using **Python**, **MJML**, and **LLM-based curation and summarization**, to study whether a bounded, non-infinite interface supports more intentional social media consumption.
- Ran a **multi-week mixed-methods field study** with **17 participants**, combining **system logs**, **weekly surveys**, and **semi-structured interviews** to characterize how users judged curation quality and what controls they wanted over the delivered set. Paper submitted to **CHI 2027**.
- **Capybara: collaborative programmable AR for children.** Helped test the mobile AR application that teaches children to code, co-ran the workshops evaluating its co-located collaboration features with **9 child participants**, and co-wrote the resulting poster paper, published at **CSCW 2026**. Currently planning a second wave of workshops toward a follow-up full paper.

Professional Experience

Surge AI **June 2026 – Present**
Product Operations Manager *New York, NY*

- Work with the team running human evaluations of **Anthropic**'s Claude models, supporting model-quality measurement for **Claude Code** testing.
- Help oversee **snapshot evaluations** (comparisons across successive Claude model versions) and **competitive evaluations** (Claude against other frontier models).
- Collect new tasks and data for evaluation prompt sets, improving the quality of the inputs models are tested on.
- Rebuild existing evaluation workflows and help incorporate new taxonomy for evaluating model behavior.
- Help onboard and guide new hires on Surge's internal tools and processes.

UnitedMasters **June 2025 – Aug. 2025**
Software Engineering Intern *Brooklyn, NY*

- Designed and shipped the Forever Saves experience within the **Masterlink** platform using **React** and **TypeScript**, letting fans permanently follow any of UnitedMasters' **1.5M+** artists on Spotify.
- Prototyped a digital impact dashboard for UnitedMasters' **20+** exclusive artists and internal teams to identify audience growth opportunities.
- Developed and deployed a pixel-tracking dashboard UI surfacing thousands of fan interactions, improving how artists interpret and act on marketing data.

Selected Projects

Few-shot Calorie-Aware Snack Detection | *Python, PyTorch, YOLOv12, Gemini API* | code **Fall 2025**

- Implemented **YOLOv1** from scratch on PASCAL-VOC and fine-tuned **YOLOv12** on a custom 7-class snack dataset, reaching **81.7% mAP@50** and **7.5% MAPE** on calorie estimation.
- Benchmarked the detector against **Gemini 2.5 Flash**, showing that explicit per-instance detection achieves **55% lower MAE** than end-to-end VLM estimation.

Orange & Blackjack | *Solidity, Web3, React, Ethers.js* | live, code **Spring 2025**

- Helped lead a five-person team as one of two developers and contributed to the conference poster and final write-up.
- Built a decentralized Blackjack game to explore how blockchain could replace the traditional casino intermediary, using Solidity smart contracts to handle LUSD betting, gameplay, and payouts with verifiable on-chain logic.
- Developed a React frontend with MetaMask wallet integration, card animations, and a live leaderboard.

Pantry 2 Plate | *Python, Flask, React, MongoDB, Selenium* | live **Fall 2024**

- Built a web application that recommends recipes based on users' pantry items and personal preferences, with customizable filters for cooking time, skill level, and dietary restrictions.
- Helped design a MongoDB schema for **6,000+** recipes and wrote custom scrapers to aggregate recipe data across multiple sources.

Leadership & Service

Business Today **July 2024 – May 2026**
Co-Director of Design Nation *Princeton, NJ*

- Co-directed **Design Nation 2026**, leading a **30-member** team across **7 sub-committees** and overseeing a **$30,000** budget to organize a national undergraduate design conference.
- Shaped conference programming and speaker selection, oversaw efforts that brought together **30+** industry leaders, and coordinated outreach to universities nationwide.

Skills

Research Methods: Mixed-Methods Design, Semi-Structured Interviews, Thematic Analysis, Survey Design, Usability Testing, Human Evaluation of LLMs
Programming & Data: Python (Pandas, NumPy, Matplotlib, Seaborn), Java, C, JavaScript/TypeScript, SQL, HTML/CSS, Solidity
Frameworks & Libraries: Flask, MJML, PyTorch, React, Scikit-learn, Selenium, Web3
Tools & Databases: Docker, Figma, GitHub, Jira, MongoDB, MySQL
Spoken Languages: English (fluent), Mandarin Chinese (fluent)

Relevant Coursework

- Computational Models of Cognition
- Machine Learning
- Computer Vision
- Intelligent Robotics
- Autonomous Fabrication
- Advanced Programming Techniques
- Algorithms & Data Structures
- Graph Theory
- Computer System Design
- Computational Biology
- Economics & Computation
- Decentralized Finance